

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 07: WORK, ENERGY & SIMPLE MACHINES

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-07

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: CONSERVATION OF MECHANICAL ENERGY IN A ROLLER COASTER LOOP

In a physics amusement park dynamics study, a model roller coaster cart of mass $m = 200 \text{ kg}$ was analyzed along a smooth vertical track (neglecting friction and air resistance):

- **Point A (Starting Hill):** Height $h_A = 20.0 \text{ m}$ above ground; cart is released from rest ($u_A = 0 \text{ m/s}$).
- **Point B (Bottom of Loop):** Ground level ($h_B = 0.0 \text{ m}$).
- **Point C (Top of Circular Loop):** Height $h_C = 15.0 \text{ m}$ above ground.
- Acceleration due to gravity $g = 9.8 \text{ m/s}^2$.

Q.1 What is the total mechanical energy ($E_{\text{mech}} = E_k + E_p$) of the cart at Point A?

- (A) 0 J (B) 39,200 J
(C) 19,600 J (D) 40,000 J

Q.2 What is the speed (v_B) of the roller coaster cart as it reaches the lowest point, Point B ($h_B = 0$)?

- (A) 14.0 m/s (B) 19.8 m/s
(C) 9.8 m/s (D) 25.0 m/s

Q.3 What is the gravitational potential energy of the cart at Point C ($h_C = 15.0 \text{ m}$)?

- (A) 29,400 J (B) 39,200 J
(C) 9,800 J (D) 14,700 J

Q.4 What is the kinetic energy (E_k) and speed (v_C) of the cart at Point C?

- (A) $E_k = 9,800 \text{ J}$; $v_C = 9.9 \text{ m/s}$ (B) $E_k = 39,200 \text{ J}$; $v_C = 19.8 \text{ m/s}$
(C) $E_k = 0 \text{ J}$; $v_C = 0 \text{ m/s}$ (D) $E_k = 19,600 \text{ J}$; $v_C = 14.0 \text{ m/s}$

Q.5 If track friction dissipated 4,200 J of energy as heat during the descent from A to B, what would the actual kinetic energy at Point B be?

- (A) 39,200 J (B) 35,000 J
(C) 43,400 J (D) 19,600 J

CASE STUDY 2: LIFTING VS DRAGGING: WORK DONE AGAINST GRAVITY AND FRICTION

A warehouse worker needs to move a heavy packing crate of mass $m = 60 \text{ kg}$ into a delivery truck whose bed is at a vertical height $h = 1.5 \text{ m}$ above the ground ($g = 9.8 \text{ m/s}^2$).

- **Method 1 (Direct Vertical Lift):** Lifting the crate straight up onto the truck bed.
- **Method 2 (Inclined Ramp):** Pushing the crate up a smooth wooden ramp of length $L = 5.0 \text{ m}$ inclined at an angle θ (where $\sin \theta = 1.5/5.0 = 0.30$). The friction force resisting motion along the ramp is $f = 80 \text{ N}$.

Q.6 What is the minimum work done by the worker against gravity in Method 1 (Direct Vertical Lift)?

- (A) 600 J (B) 882 J
(C) 90 J (D) 450 J

Q.7 In Method 2, what is the work done solely to overcome friction (W_{friction}) along the 5.0 m ramp?

- (A) 400 J (B) 80 J
(C) 882 J (D) 120 J

Q.8 What is the total work done by the worker in pushing the crate up the ramp in Method 2?

- (A) 882 J (B) 1282 J
(C) 400 J (D) 482 J

Q.9 If Method 2 requires more total energy (1282 J vs 882 J), why does the worker choose to use the ramp?

- (A) Because it saves electricity (B) Because the ramp reduces the required muscular pushing effort (force) by spreading the work over a longer distance
(C) Because gravity is zero on a ramp (D) Because it takes less time

Q.10 What is the mechanical efficiency (η) of the inclined ramp as a machine in Method 2?

- (A) 100% (B) 68.8%
(C) 45.2% (D) 31.2%

CASE STUDY 3: ELECTRIC MOTOR POWER & OVERHEAD WATER PUMPING EFFICIENCY

A residential building uses an electric water pump to fill an overhead water storage tank on the roof.

- Tank capacity: $V = 1000 \text{ Liters} = 1.0 \text{ m}^3$ (mass of water $m = 1000 \text{ kg}$).
- Vertical height of the tank above the ground sump: $h = 18.0 \text{ m}$.
- Time taken by the pump to fill the tank completely: $t = 5 \text{ minutes} = 300 \text{ seconds}$.
- Electric power rating of the pump motor: $P_{\text{in}} = 1000 \text{ W} = 1.0 \text{ kW}$.
- Acceleration due to gravity $g = 9.8 \text{ m/s}^2$.

Q.11 What is the useful mechanical work output ($\$W_{\text{out}}\$$) performed by the pump in lifting the 1000 kg of water?

- (A) 18,000 J (B) 176,400 J
(C) 180,000 J (D) 300,000 J

Q.12 What is the useful mechanical power output ($\$P_{\text{out}}\$$) of the pump during the operation?

- (A) 588 W (B) 1000 W
(C) 352.8 W (D) 176.4 W

Q.13 What is the total electrical energy consumed by the 1000 W motor in 5 minutes (300 s)?

- (A) 300,000 J (B) 176,400 J
(C) 50,000 J (D) 600,000 J

Q.14 Calculate the percentage efficiency (η) of the water pumping system:

- (A) 100% (B) 58.8%
(C) 72.4% (D) 85.0%

Q.15 How many commercial units of electrical energy (kilowatt-hours, kWh) are consumed by the pump during this 5-minute run?

- (A) 0.083 kWh (B) 1.0 kWh
(C) 0.5 kWh (D) 0.05 kWh

CASE STUDY 4: PRINCIPLE OF MOMENTS & MECHANICAL ADVANTAGE OF LEVERS

Students conducted an experiment on a 1.00 m uniform meter stick pivoted at its center (fulcrum $\$F\$$ at the 50.0 cm mark) to study levers and the principle of moments.

• Class 1 Lever Investigation:

- A load of mass $\$M = 300\text{ g}\$$ ($\$W = 3.0\text{ N}\$$) was hung at the 10.0 cm mark (Load arm $\$d_L = 50.0 - 10.0 = 40.0\text{ cm}\$$).

- An effort force was applied by hanging slotted masses at the 90.0 cm mark (Effort arm $\$d_E = 90.0 - 50.0 = 40.0\text{ cm}\$$) to restore rotational equilibrium.

• Next, the effort was moved to the 100.0 cm mark ($\$d_E = 50.0\text{ cm}\$$) while keeping the load fixed at 10.0 cm.

Q.16 According to the Principle of Moments, what condition governs rotational equilibrium?

- (A) Total clockwise moment equals total anticlockwise moment about the fulcrum (B) Mass on the right must be larger than mass on the left
(C) Velocity ratio must equal zero (D) Effort force must equal zero

Q.17 What effort force ($\$E\$$) is required to balance the 3.0 N load when the effort is at the 100.0 cm mark ($\$d_E = 50.0\text{ cm}\$, \$d_L = 40.0\text{ cm}\$$)?

- (A) 3.0 N (B) 2.4 N
(C) 1.5 N (D) 4.0 N

Q.18 What is the Mechanical Advantage ($\$ \text{ext}\{MA\} = \text{ext}\{Load\} / \text{ext}\{Effort\} \$$) of this lever when $\$E = 2.4\text{ N}\$$ and $\$ \text{ext}\{Load\} = 3.0\text{ N}\$$?

- (A) 1.0 (B) 1.25
(C) 0.80 (D) 2.0

Q.19 Which common tool operates as a Class 2 lever where the load lies between the fulcrum and the effort?

- (A) A pair of scissors (B) A wheelbarrow or nutcracker
(C) A pair of laboratory tweezers (D) A crowbar

Q.20 Why is the mechanical advantage of a Class 3 lever (such as a fishing rod or human forearm) always less than 1?

- (A) Because the effort arm is shorter than the load arm ($d_E < d_L$), sacrificing force to gain speed and displacement (B) Because of friction
(C) Because energy is destroyed (D) Because it is broken

CASE STUDY 5: MECHANICAL ADVANTAGE & EFFICIENCY OF PULLEY SYSTEMS

Students set up two pulley arrangements to lift a heavy block of mass $m = 20.0 \text{ kg}$ (Load $W = 196 \text{ N}$):

- **System A (Single Fixed Pulley):** A single pulley fixed to an overhead rigid ceiling support. A student pulls downward on the rope to raise the load vertically upward.
- **System B (Block and Tackle with 4 Pulleys):** Two pulleys in the upper fixed block and two pulleys in the lower movable block. The load is suspended from the lower block. The total mass of the lower block and hooks is $w_{\text{pulley}} = 2.0 \text{ kg}$ (19.6 N).

Q.21 What is the Velocity Ratio (VR) and Ideal Mechanical Advantage of System A (Single Fixed Pulley)?

- (A) 1 (B) 2
(C) 4 (D) 0.5

Q.22 What is the primary mechanical advantage of using a single fixed pulley if it does not reduce the required effort force?

- (A) It creates energy (B) It allows effort to be applied in a convenient downward direction (using the worker's own body weight) rather than lifting upward
(C) It multiplies velocity by 10 (D) It eliminates gravity

Q.23 In System B (Block and Tackle with 4 pulleys), how many parallel rope segments support the movable load block?

- (A) 2 (B) 4
(C) 6 (D) 1

Q.24 What ideal effort (E) is required in System B to lift the 196 N load if pulley weight and friction are neglected?

- (A) 196 N (B) 98 N
(C) 49 N (D) 24.5 N

Q.25 If the lower movable pulley block has a weight of 19.6 N, what actual effort (E_{actual}) is required (neglecting axle friction)?

- (A) 49 N (B) 53.9 N
(C) 98 N (D) 44.1 N

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	$E_{\text{mech}} = mgh_A + \frac{1}{2}m u_A^2 = 200 \text{ kg} \times 9.8 \text{ m/s}^2 \times 20.0 \text{ m} + 0 = 39,200 \text{ J}$.
Q.2	(B)	By conservation of energy: $\frac{1}{2}m v_B^2 = mgh_A \Rightarrow v_B = \sqrt{2gh_A} = \sqrt{2 \times 9.8 \times 20.0} = \sqrt{392} \approx 19.8 \text{ m/s}$.
Q.3	(A)	$E_p = mgh_C = 200 \times 9.8 \times 15.0 = 29,400 \text{ J}$.
Q.4	(A)	$E_{k, C} = E_{\text{total}} - E_{p, C} = 39,200 - 29,400 = 9,800 \text{ J}$. Speed: $v_C = \sqrt{\frac{2 E_{k, C}}{m}} = \sqrt{\frac{2 \times 9800}{200}} = \sqrt{98} \approx 9.9 \text{ m/s}$.
Q.5	(B)	Actual $E_k = E_{\text{initial}} - E_{\text{friction}} = 39,200 \text{ J} - 4,200 \text{ J} = 35,000 \text{ J}$.
Q.6	(B)	$W_{\text{gravity}} = mgh = 60 \text{ kg} \times 9.8 \text{ m/s}^2 \times 1.5 \text{ m} = 882 \text{ J}$.
Q.7	(A)	$W_{\text{friction}} = f \times L = 80 \text{ N} \times 5.0 \text{ m} = 400 \text{ J}$.
Q.8	(B)	$W_{\text{total}} = W_{\text{gravity}} + W_{\text{friction}} = 882 \text{ J} + 400 \text{ J} = 1282 \text{ J}$.
Q.9	(B)	Simple machines act as force multipliers: Lifting requires 588 N , while pushing up the ramp requires only $F = mg \sin \theta + f = (60 \times 9.8 \times 0.30) + 80 = 176.4 + 80 = 256.4 \text{ N}$.
Q.10	(B)	$\text{Efficiency} = \frac{\text{Useful Work Output (against gravity)}}{\text{Total Work Input}} \times 100 = \frac{882 \text{ J}}{1282 \text{ J}} \times 100 \approx 68.8\%$.
Q.11	(B)	$W_{\text{out}} = mgh = 1000 \text{ kg} \times 9.8 \text{ m/s}^2 \times 18.0 \text{ m} = 176,400 \text{ J}$ (176.4 kJ).
Q.12	(A)	$P_{\text{out}} = \frac{W_{\text{out}}}{t} = \frac{176,400 \text{ J}}{300 \text{ s}} = 588 \text{ W}$.
Q.13	(A)	$E_{\text{consumed}} = P_{\text{in}} \times t = 1000 \text{ W} \times 300 \text{ s} = 300,000 \text{ J} = 300 \text{ kJ}$.
Q.14	(B)	$\text{Efficiency} = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100 = \frac{588 \text{ W}}{1000 \text{ W}} \times 100 = 58.8\%$.
Q.15	(A)	Energy in $\text{kWh} = \text{Power (kW)} \times \text{Time (hours)} = 1.0 \text{ kW} \times \left(\frac{5}{60}\right) \text{ h} = \frac{1}{12} \text{ kWh} \approx 0.0833 \text{ kWh}$.
Q.16	(A)	A lever is in rotational equilibrium when the algebraic sum of all moments about the pivot equals zero: $\text{Anticlockwise Moment} = \text{Clockwise Moment}$.
Q.17	(B)	$\text{Effort} \times d_E = \text{Load} \times d_L \Rightarrow E \times 50.0 = 3.0 \times 40.0 \Rightarrow E = \frac{120}{50} = 2.4 \text{ N}$.
Q.18	(B)	$\text{MA} = \frac{\text{Load}}{\text{Effort}} = \frac{3.0 \text{ N}}{2.4 \text{ N}} = 1.25$ (or $\text{MA} = \frac{d_E}{d_L} = \frac{50}{40} = 1.25$).
Q.19	(B)	In a wheelbarrow and nutcracker, the load is in the middle (between fulcrum and effort), giving a mechanical advantage always greater than 1.
Q.20	(A)	In Class 3 levers, the effort is applied between the fulcrum and load ($d_E < d_L$). While $\text{MA} < 1$ requires more force, it multiplies speed and displacement.
Q.21	(A)	In a single fixed pulley, the effort moves the exact same distance as the load ($\text{VR} = 1$). It changes the direction of effort without multiplying force.
Q.22	(B)	Pulling downward allows the worker to utilize body weight effectively and ergonomically to raise loads upward.
Q.23	(B)	In a 4-pulley system (2 fixed, 2 movable), the load is shared equally among $n = 4$ supporting segments of the continuous rope, making $\text{VR} = 4$.
Q.24	(C)	$E = \frac{\text{Load}}{n} = \frac{196 \text{ N}}{4} = 49 \text{ N}$.

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.25	(B)	Total weight lifted $= \text{Load} + w_{\text{pulley}} = 196 + 19.6 = 215.6 \text{ N}$. Actual effort $= \frac{215.6}{4} = 53.9 \text{ N}$.